Blood flow measurement using µPIV: how accurate really is it?

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Abstract. Particle image velocimetry (PIV) is a widely used technique in science and engineering, allowing for non-invasive, real-time measurement of flow velocities. The measurements, however, are often prone to significant systematic errors and uncertainties which are rarely considered in biological contexts. In this project a PIV system was used to measure blood flow velocities within the zebrafish heart. A new method was developed to quantify the uncertainties on correlation-averaged PIV measurements. The technique was evaluated using synthetic videos and found to produce good estimates, generally only producing a small overestimate of their magnitude. The technique did, however, produce some significant underestimates. Possible reasons for this are discussed. The uncertainty quantification method was applied to the zebrafish heart, where it was found that there is significant uncertainty in the valve region when flow velocities are high. Finally, the key assumption that there exists limited beat-to-beat variability within the zebrafish heart was tested using a variety of techniques. These techniques were tested on synthetic videos to evaluate their efficacy. An adapted form of the modified z-score method showed promise in detecting beat-to-beat variability. Applying the method to the zebrafish heart found no strong evidence of significant beat-to-beat variability indicating that the underlying assumptions of the method are valid.

1. Introduction

The measurement of fluid velocities is an essential technique in science and engineering, allowing for the verification of mathematical models, and providing new insight into physical and biological systems [1]. Particle image velocimetry (PIV) is a method, pioneered in 1984 [2], employed for this purpose which allows for non-invasive, real-time measurement of flow velocities. When applied to biological systems the technique is however often subject to large systematic errors and statistical uncertainties and, despite the long history of the method, there has been limited research on quantifying these. The zebrafish heart offers an ideal model for investigating this in a biological context: embryonic zebrafish are transparent and through genetic modification the red blood cells can be made to fluoresce, avoiding the need to introduce tracers which may disrupt blood flow.

Zickus and Taylor [3] previously used PIV to measure blood flow velocities within the zebrafish heart, applying a correlation-averaging technique to overcome the low seeding densities. This correlation averaging technique makes uncertainty quantification difficult as it only produces a

single measurement meaning an estimate of the distribution of velocities cannot be made directly. Additionally, the method assumes that there exists minimal beat-to-beat variability and that the phase-matching technique [4] works well. This project recreates and builds on their work - introducing a new method to quantify the uncertainties on these measurements and applying methods to evaluate the validity of these assumptions.

In the first part of this project the blood flow velocities in the zebrafish heart are measured using a PIV system. Phase-binning and correlation averaging are used to improve the accuracy of the measurements under the low-seeding density found. A new approach is then introduced which allows for the uncertainties to be quantified in correlation-averaged PIV. This approach uses bootstrapping to produce new samples from the correlation matrices, providing an estimate of the underlying distribution. Synthetic videos are used to evaluate the method and the approach is then extended to evaluate the phase-binning technique to investigate whether there is significant temporal velocity gradients across phase bins. Finally the bootstrapping, bootlier, modified z-score and, a new image-transformation technique are applied to identify any significant beat-to-beat variability within the heart. The approaches are evaluated using synthetic images.

2. Background

2.1. Particle image velocimetry

Particle image velocimetry is a technique which applies pattern matching to measure flow velocities within fluids. The approach uses recordings of fluids seeded with tracer particles under the assumption that these tracer particles reliably follow the motion of the fluid [5]. Under ideal conditions the tracer particles also should not alter the flow or interact with each other [5]. The method works by defining an interrogation window around the region of interest in a given frame. Next, in the subsequent frame, the interrogation window from the first frame is translated over a search region calculating a similarity index for each possible location. The values of the similarity index are saved in a 2-dimensional array: the correlation matrix. The location in the correlation matrix where the similarity index is maximised, relative to the centre of the interrogation window, is the estimated displacement between the two frames for this location. Under the assumption that the spacial acceleration within the window is small, the average velocity within the window, $\vec{u}$, can then be calculated using the formula:

$$\vec{u} = M\frac{\Delta\vec{x}}{\Delta t}, \tag{1}$$

where $\Delta\vec{x}$ is the displacement vector, Δt is the time between frames and M is a scaling factor which relates pixel dimensions and real-world dimensions.

2.1.1. Correlation averaging

For measuring time-averaged flow, the individual velocity measurements can be averaged. This however can produce erroneous measurements when seeding density is low [6]. Correlation-averaging is a technique which can substantially improve measurements under these conditions [6]. The technique takes each individual correlation matrix over multiple measurements and averages them pixel-by-pixel producing a new correlation-averaged matrix. The displacement can then be estimated by finding the location which maximises the similarity index in this matrix. Figure 1 shows the correlation averaging technique applied to a synthetic video with low seeding density and significant noise.

2.1.2. Sub-pixel interpolation

To allow for non-integer displacements a sub-pixel interpolation method is needed. Figure 2 shows the application of a 5-point Gaussian interpolation in 1-dimension. Most often a parabolic,

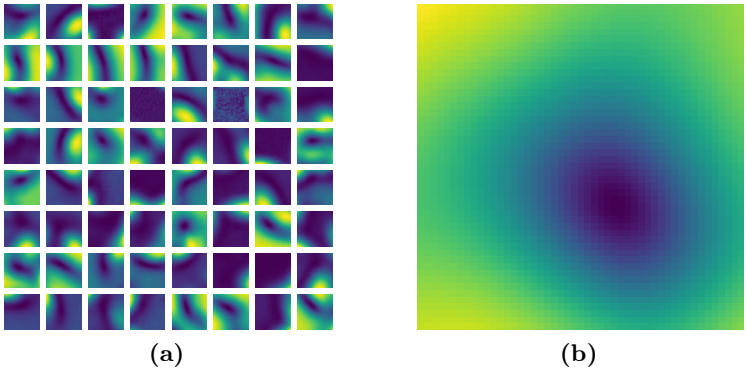


Figure 1: Plot showing (a) normalised individual frame pair correlation matrices and (b) correlation matrix produced from correlation averaging these correlation matrices. Correlation matrices which contain no particles produce erroneous displacement estimates. The correlation averaged measurement suppresses these producing a single localised estimate of the displacement.

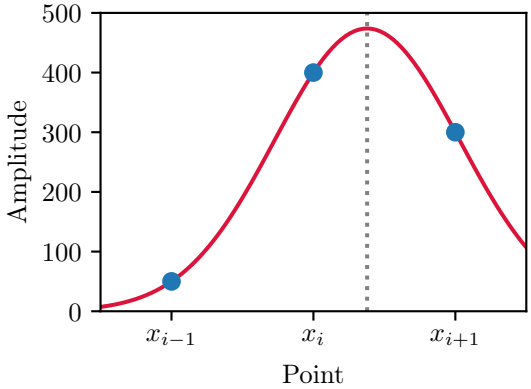


Figure 2: Gaussian interpolation in 1-dimension. Application of this technique in 2-dimensions around the peak location allows for sub-pixel displacements to be determined.

local 5-point Gaussian, or a sinc interpolation method is used [7]. The approaches used in this project make these unsuitable due to a significant peak-locking bias, as such a 9-point Gaussian interpolation method [8] was used.

2.1.3. Window offsetting

Care must be taken to select the optimal sizes for the interrogation window and search region. if too large an interrogation window is selected, then the spatial resolution is reduced and the assumption that the velocity gradients are small becomes increasingly unlikely. Alternatively, if too small an interrogation window is used the result is greater levels of noise and therefore higher measurement uncertainties. An additional consideration is that if the velocities in the region of interest are high and seeding densities are low then particles may enter the interrogation window in the second frame [1]. The probability of this is increased with larger interrogation window sizes [1]. This is known as loss of correlation [1]. One solution to this is using a large search region but this results in greater statistical uncertainty [1]. Westerwheel proposed a solution to this known

as discrete window-offsetting [9]. Discrete window-offsetting is a multi-pass PIV technique which allows for increased spatial resolution and a reduction in noise [9]. The method first applies the standard PIV technique as described above, then for a second pass the search region is offset by the displacement calculated in the first pass [9]. The technique allows for a smaller search region in the second pass eliminating this bias and resulting in reduced noise [9].

2.1.4. Uncertainty quantification in particle image velocimetry

There are a large number of uncertainties in PIV, both systematic and random in nature. These uncertainties originate from the image acquisition technique and from the method used to determine the flow vectors [10]. The uncertainty on a given velocity in PIV can be derived from Equation (1) giving:

$$\sigma_u = u\sqrt{\frac{\sigma_M}{M}^2 + \frac{\sigma_x}{x}^2 + \frac{\sigma_t}{t}^2},$$

where the σ subscripts represent the uncertainty on the given value. The uncertainties on M and t can be obtained prior to the experiment which just leaves the uncertainty on x.

Non-systematic uncertainties would normally be estimated by obtaining multiple measurements and calculating the standard error. The correlation-averaging method, however, only produces a single value for a given region and therefore standard statistical formulae for standard error are unsuitable, motivating the need for an alternative approach. A number of techniques have been suggested in literature to estimate the uncertainty [11][12][13][14] but they often rely on empirical relations which require assumptions about the PIV system being used. This project applies a bootstrapping approach to the problem. Bootstrapping is a set of related Monte-Carlo statistical techniques which allow for the estimation of the distribution of a population using a smaller sub-sample [15]. The method works under the assumption that the data-set is a representative sample of the underlying distribution. Bootstrapping uses sampling with replacement to form new sets of data for which we can then use to calculate a desired statistic. Collecting large sets of these samples allows us to determine the distribution of said statistic.

2.1.5. Particle image velocimetry in the zebrafish heart

The zebrafish heart is an ideal organism for measuring flow velocities in a biological context. Major advantages include: the transparency of the heart making imaging easier; at smaller scales the Reynolds number is small resulting in low levels of turbulence [16]; and the widespread use of zebrafish in biological studies.

2.2. Single-plane illumination microscopy

The small scale of the zebrafish heart requires the use of micro-PIV (µPIV). Limited optical access means that µPIV systems must use volume illumination meaning that scattered light can make illuminated particles more difficult to image [17]. We can reduce scattering by using fluorescent particles and single-plane illumination microscopy [17]. Single-plane illumination microscopy (SPIM) is an imaging technique which utilises a light sheet to selectively illuminate a narrow slice of an object. This limits the additional systematic uncertainty caused by imaging outside the focal plane of the lens known as depth of correlation [17].

2.3. Identifying intra-individual variability

Intra-individual variability refers to variations found within a single individual. It can be difficult to separate these from measurement error. In this project the modified z-score [18] and the bootlier method [19] are applied to in an attempt to identify these.

The z-score approach calculates how many standard deviations from the mean a measurement lies where a z-score which is ≥ 3 indicates that a measurement is an outlier [18]. The technique does not work well for smaller data sets [18]. An alternative is the modified z-score, M_i, given by the equation:

$$M_i = \frac{0.6745(x - \tilde{x})}{MAD}, \tag{2}$$

where x_i is the value of a measurement, $\tilde{x}$ is the median of all measurements and MAD is the median of absolute differences between each measurement and the median [18]. The approach performs better than the z-score because the median is less sensitive to outliers [18].

Visual inspection of the bootstrap distribution would show outliers as additional peaks. The bootstrapping method, however, is not highly sensitive to multimodality as the peaks tend to be smoothed out in the process [19]. To improve its sensitivity Singh proposed the bootlier approach [19]. In this method, for each bootstrapped sample the two highest and lowest measurements are removed. The mean is then calculated and subtracted from the mean of the original sample. These values are then plotted a histogram where each peak represents an outlier and unimodality would indicate that there are no outliers.

In this project we additionally propose another method which works under the assumption that data without intra-individual variability will be normally distributed and the presence of variability will result in a non-normal distribution. This non-normality can be quantified this by calculating the Fisher-Pearson coefficient of skew [20] and the Fisher definition of kurtosis [20] of the bootstrapped distribution.

3. Methods and analysis

3.1. PIV system

The PIV system used is described in Section 2.1. The system uses the sum of absolute differences metric as a similarity index where the minimum of this gives the peak location. Due to significant peak-locking in the uncertainty quantification process, a Gaussian filter with a 3 px standard deviation was applied to the correlation matrices before estimating the displacement.

3.2. Acquisition of zebrafish heart video

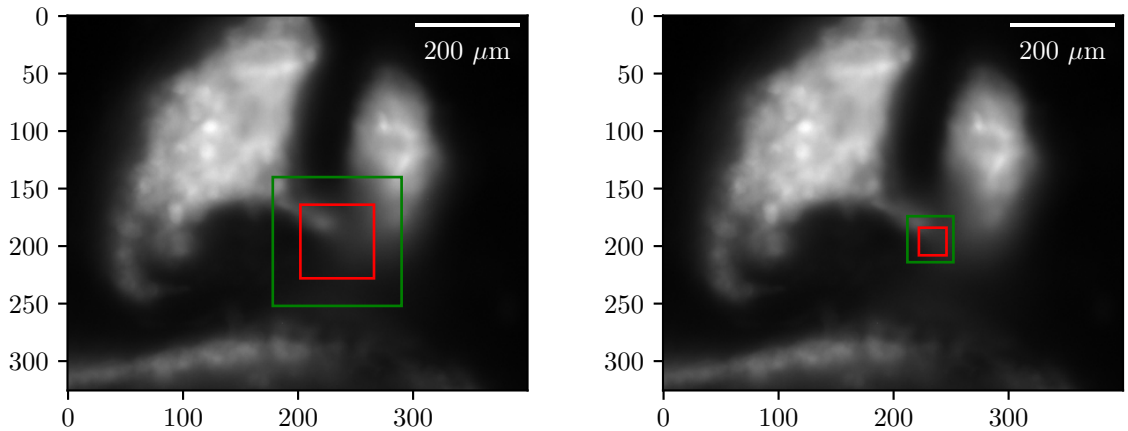
Phase-binned zebrafish video data was provided by Dr Jonathan Taylor. A brief description of the method is given here for completeness but full details are given in [3]. A brightfield video for phase identification and a SPIM video for PIV analysis of a embryonic zebrafish heart were recorded simultaneously. The phase of the heart was determined by finding the minimum sum of absolute differences between each frame and the subsequent frames. An interpolation was then applied to assign each SPIM frame a phase between 0 and 2π.

3.3. Measurement of blood flow velocities in the zebrafish heart

The SPIM videos were split into pairs of consecutive frames and subsequently binned into 30 equal-sized phase bins. PIV analysis was performed at x and y coordinates of 234 px and 196 px respectively. Two passes were performed: first with a 64 px interrogation window and a 24 px search region and then with a 24 px interrogation window and an 8 px search region. The analysis region is shown in Figure 3. Both the correlation-averaged estimates of the x-displacement and the non-correlation-averaged estimates of the x-displacements were saved with their corresponding phases.

3.4. Uncertainty quantification on blood flow velocities in the zebrafish heart

A bootstrapping approach was employed to estimate the uncertainty on correlation-averaged measurements; for each phase-bin the correlation matrices were resampled with replacement to



(a) **(b)**

Figure 3: Analysis region for the zebrafish heart at a phase between 2.93 rad and 3.12 rad where (a) shows the first pass region and (b) shows the second pass region. The left region is the atrium and on the right, connected by the valve region, is the ventricle. The red box shows the interrogation window and the green box the search region. The two-pass PIV algorithm allows for greater spatial resolution and reduced uncertainty.

produce new samples equal in size to the original sample. The displacement vector was then estimated and saved; this was repeated 1,000 times. The standard deviation of the displacement vector was then calculated which gave an estimate of the standard error on the mean displacement for each phase. The non-correlation averaged and correlation averaged measurements with their uncertainties were plotted against phase.

3.5. Evaluation of uncertainty quantification method

To validate the method of quantifying uncertainty and further evaluate the PIV measurements, a system was developed to produce synthetic videos. These videos consisted of 16-bit pairs of frames of arbitrary dimensions and a specified number of frames. Each frame pair contained particles which were placed at random locations in a given region. The particles were each modelled by a 2-dimensional, rotated, elongated Gaussian with equation:

$$I = B \exp\left(\frac{-x'^2}{2\sigma_x^2} + \frac{-y'^2}{2 \cdot 1.2 \cdot \sigma_y^2}\right),$$

$$\begin{aligned} \text{where } x &= x_i - x_\mu, \\ y &= y_i - y_\mu, \\ x' &= x\cos\theta - y\sin\theta, \\ y' &= x\sin\theta + y\sin\theta, \end{aligned}$$

I is the pixel value, x_i and y_i is the pixel location, x_μ and y_μ are the particle centroids, θ is the rotation of the particle, B is the particle intensity and, σ_x and σ_y are the particle sizes. The particle intensity was additionally sampled from a normal distribution with user-defined mean

and standard deviations. Between frame pairs, the particles were displaced by a distance defined by a function and the particle intensity was resampled from its initial distribution. Additionally, noise from a normal distribution with mean 0.0 and a standard deviation of 0.5 was added to all frames.

2,500 videos of dimensions 72×72 px were produced consisting of 78 frame-pairs each containing a single particle of particle intensity of 8 ± 1 and a size of 14 px. The particles were initially placed in a 96×96 px region for the first frame in a frame pair. The particles were then translated, with $x(t)$ and $y(t)$ the current coordinates and $x(t+1)$ and $y(t+1)$ the coordinates in the subsequent frame, by the functions:

$$x(t+1) = x(t) + 3,$$
$$y(t+1) = y(t) + 3.$$

The correlation-averaged x-displacement vector, v_i was calculated for each of these videos using a single PIV pass in the centre of the frame with an interrogation window of 44 px and a search region of 8 px. The standard deviation, σ_μ, of these x-displacement vectors was calculated giving an estimate of the standard error on the mean. The standard error on the x-displacement, σ_i, was also calculated separately for each video using the bootstrapping approach with a sample size of 1,000. The mean of the standard errors calculated using the first method was compared to the standard error calculated using the bootstrapping method. The uncertainties on these estimates were additionally quantified. For the first method a bootstrapping approach was applied again, resampling the standard errors 2,500 times and taking the standard deviation to estimate the standard error. For the second approach the formula for standard error was used. For visual confirmation that the bootstrapped distributions were Gaussian a single plot of the bootstrapped distribution was saved with a sample size of 1,000. The following formula was then used to calculate the studentised residual, s, for the bootstrapped standard errors:

$$s = \frac{v_i - \sum_{i=1}^{n} v_i}{n\sigma_i},$$

where $n = 2,500$ is the number of videos. The values of s were then plotted on a histogram with a t-distribution with $n-1$ degrees of freedom. The expectation was that the histogram distribution would match the t-distribution if the uncertainty quantification method is robust. The t-distribution can be approximated to a normal distribution so the null hypothesis that the distribution of s comes from a normal distribution was tested using Scipy's [21] implementation of D'Agostino's and Pearsons test [22][23]. The skew, kurtosis, and standard deviation were also calculated for the distribution as measures of normality. A skew and kurtosis close to zero and a standard deviation close to one would indicate that the distribution is close to a normal distribution. To estimate the uncertainty on the estimates of skew, kurtosis, and standard deviation the distribution was then resampled with replacement 1000 times and the skew, kurtosis, and standard deviation, were calculated for each sample. The standard deviation of these measurements was taken to give an estimate on their associated uncertainties.

3.6. Evaluation of phase-binning and correlation averaging

The bootstrap approach was applied again for the phase bin between 2.93 rad and 3.12 rad at x and y coordinates of 196 px and 234 px respectively using the same PIV settings as in Section 3.3. The estimated x-displacement, the mean phase, timestamp, and non-correlation averaged displacement were saved. Plots of the correlation-averaged mean against mean phase, timestamp and non-correlation averaged displacement were saved.

3.7. Identifying beat-to-beat variability within the zebrafish heart

Multiple approaches were applied to identify beat-to-beat variability. These were evaluated using two sets of 1,000 videos with and without the presence of outliers. The videos were of dimensions 72×72 px and consisted of 78 frame-pairs each with a particle intensity of 8 ± 1 and a size of 14 px. The functions defining their motion, with $x(t)$ and $y(t)$ the current coordinates, $x(t+1)$ and $y(t+1)$ the coordinates in the subsequent frame was:

$$x(t+1) = \begin{cases} x(t) - 3, & \text{if } r \geq 0.1 \\ x(t) + 3, & \text{otherwise,} \end{cases}$$

$$y(t+1) = \begin{cases} y(t) - 3, & \text{if } r \geq 0.1 \\ y(t) + 3, & \text{otherwise.} \end{cases}$$

The value of r was sampled from a uniform distribution between 0 and 1 for the first set of videos and fixed at 1 for the second set. This means that the first set of videos has on average 10% of its frames containing outliers and the second set contains no outliers.

3.7.1. Bootstrapping

To identify beat-to-beat variability in the zebrafish heart the skew and kurtosis of the bootstrap distribution were calculated for the phase bin between 2.93 rad and 3.12 rad. 50,000 bootstrap iterations were used and then 1,000 repetitions were used to get an estimate of the standard error on the skew and kurtosis. The analysis was done at x and y coordinates of 196 px and 234 px respectively. Two passes were performed, first with a 64 px interrogation window and a 24 px search region and then with a 24 px interrogation window and a 8 px search region. The distribution was also visually inspected for multimodality.

To validate the approach the two sets of synthetic videos with and without outliers were individually analysed. Due to the small size of the videos only a single pass was used with an interrogation window of 24 px and a search region of 12 px. For each video the bootstrap distribution was calculated with 100 iterations and the skew and kurtosis were calculated. The mean and standard error for each these was then calculated and the two sets were compared. Student t-test's were then performed on the distribution of skew and kurtosis comparing the two sets of videos. The alternative hypothesis that the set of videos with outliers would have greater mean skew and kurtosis was tested.

3.7.2. Bootlier

The bootlier approach was applied to the phase-binned correlation matrices at phases between 2.93 rad and 3.12 rad. The analysis was performed at x and y coordinates of 196 px and 234 px respectively. The interrogation window size was set to 24 px and the search region to 12 px. The bootstrap was performed with 100 resamples. For the set of resamples the two highest and lowest x-displacement measurements were removed and the mean was then calculated and saved. This was repeated 5000 times and the means were plotted on a histogram. The approach was repeated for two of the synthetic videos with the same PIV settings and the histograms were saved.

3.7.3. Modified Z-score

The bootstrap approach was applied again to the phase-binned zebrafish correlation matrices with 50000 iterations. The PIV system used a 24 px interrogation window and a 12 px search region. The modified z-score, given by Equation (2), was then calculated for each bootstrap measurement. The presence of outliers was then quantified by finding measurements with a modified z-score greater than 3.5 or less than -3.5. This was repeated for the synthetic videos with and without

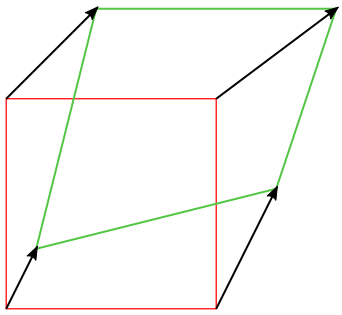


Figure 4: Window-transformation method. The original image shown in red is mapped onto the green region defined by the displacement vectors estimated at the vertices of the red square. The technique provides a method to visually identify beat-to-beat variability and validate PIV measurements.

outliers and the mean number of outliers identified was saved. The two sets of videos were then compared with a t-test under the alternative hypothesis that the set of videos with outliers has a greater value than the set of videos without outliers.

3.7.4. Window-transformation

A image transformation technique was used to identify beat-to-beat variability and qualitatively evaluate the PIV system's performance. First evenly spaced locations with spacing of 24 px in the zebrafish heart were defined. At each location the PIV system was used to estimate the velocity using a 24 px interrogation window and a 24 px search region. The uncertainty was then calculated using 100 bootstrap iterations. Two meshes were then defined: the first mesh used the evenly spaced coordinates, and the second added the estimated velocity to each of these coordinates plus and minus the uncertainty. The first frame in each image pair was then transformed from the first mesh onto the mesh with the lower and upper bound uncertainties. This gives us an estimate of what the second frame should look like if our displacement estimates are accurate. Figure 4 shows the transformation for a single rectangle. The transformed frames were compared to the non-transformed frames for the phase bin between 2.93 rad and 3.12 rad.

4. Results and discussion

4.1. Measurement of blood flow velocities in the zebrafish heart

Figure 5 shows the distribution of velocities at a single location within the zebrafish heart. The region around 5 rad shows high uncertainties in both the correlation-averaged and non-correlation-averaged measurements. Additionally we see that in this region there is some deviation between the two methods. The spread of the non-correlation averaged-measurements suggests this could be due to low seeding density which produces a uniformly distributed range of velocities. Adding this to the frames which contain particles produces a systematic zero-bias on the measurements. This provides good justification of the use of the correlation-averaging method. The uncertainties on the two methods show some level of agreement providing some verification of the uncertainty quantification method.

4.2. Evaluation of uncertainty quantification method

The studentised residuals distribution is shown in Figure 6b. The expectation was that if the uncertainty quantification method was robust then this would closely match the t-distribution with 2,499 degrees of freedom. As the degrees of freedom were relatively high we can approximate this to a normal distribution. A visual inspection suggests a close match but the p-value for the

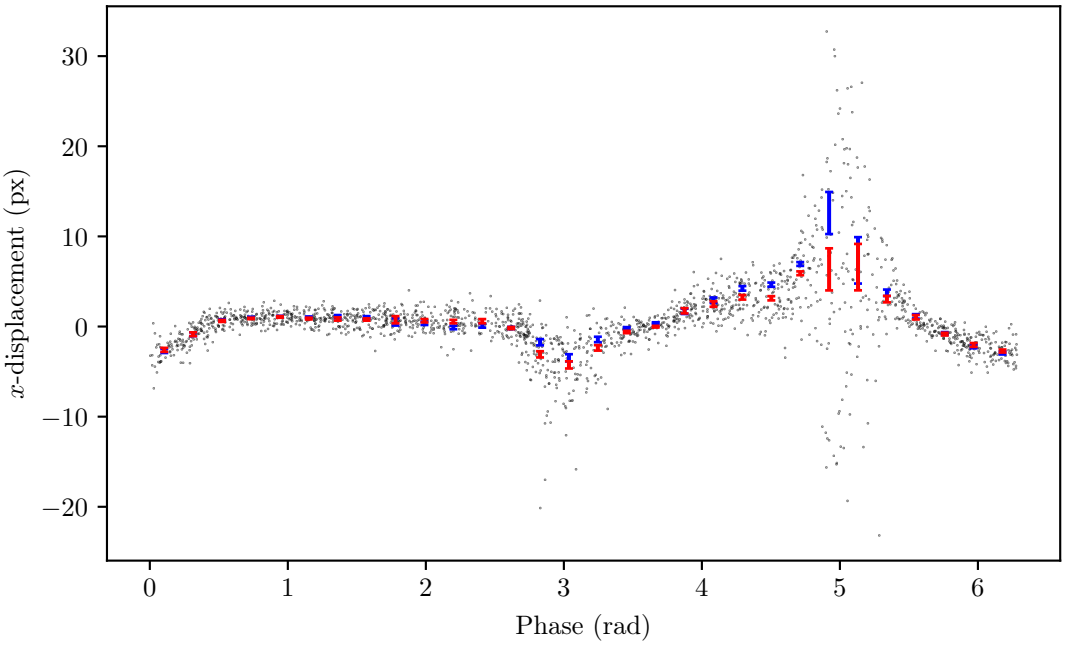
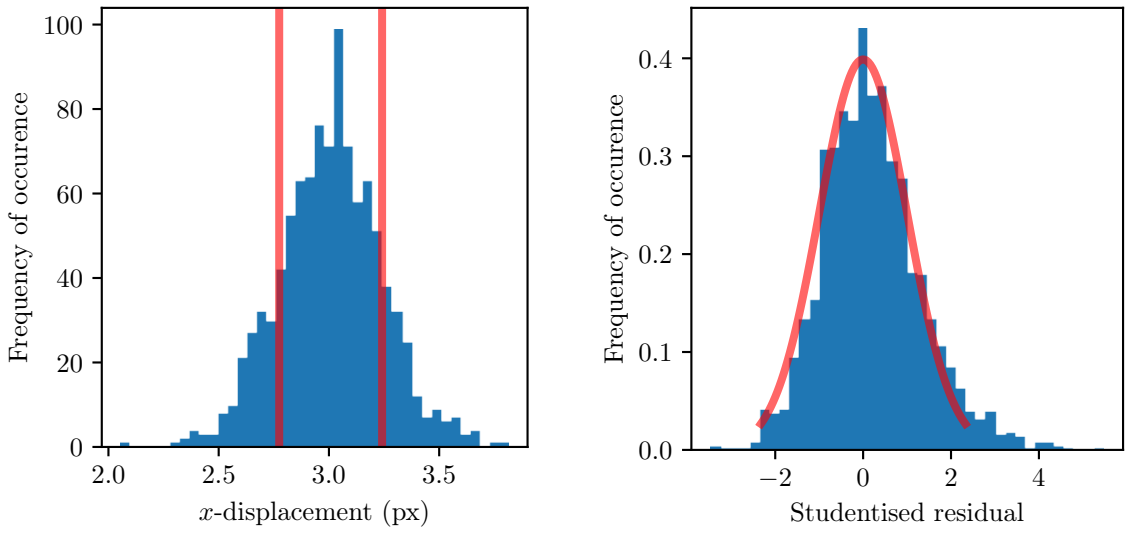


Figure 5: Zebrafish flow at fixed location. Non-correlation averaged results are shown in black with the mean and standard deviation shown in red. Correlation averaged results and their associated uncertainties are shown in blue. The non-correlation-averaged results show significant spread at around 3 and 5 rad.

null hypothesis that the distribution was from a normal distribution was 6.41×10^{-31}. This is a very low p-value indicating, at a very high confidence level, that the distribution is not normally distributed. Indeed a quantitative analysis finds that the distribution shows a significant kurtosis of 0.83 ± 0.21 suggesting the method overestimates the uncertainty. The standard deviation of the distribution of 1.13 ± 0.02 also suggests there is some overestimation of the uncertainty. On the other hand, the skew of the distribution is 0.21 ± 0.07 meaning the distribution is skewed in the direction of the displacement. This suggests there are some outliers on the right side of the distribution, i.e there is a small probability that some uncertainty estimates are highly underestimated. We can see these on Figure 6b where there are studentised residuals greater than 5, which, with a sample size of 2,500 are unlikely down to chance.

The deviation from the expected distribution could partially due to the assumption that the velocities follow a normal-distribution and therefore calculating the percentiles of the bootstrapped distribution may give better results. Figure 6a does appear Gaussian but it may be that other videos, particularly those with a significant number of noise-frames have a different distribution. Additionally, it could be that the sample size of 78 was too small and not a representative sample. This sample size was chosen as it was roughly the number of frame-pairs found in the zebrafish data. This suggests a larger bin size may be required for the zebrafish analysis to accurately estimate the uncertainty.

Despite these limitations the bootstrapped standard error was 0.372 ± 0.002 compared to the mean standard deviation of the videos of 0.354 ± 0.006. This suggests the method works remarkably well, with only a small bias towards overestimating the uncertainty. The method also offers many advantages over alternatives which often require additional assumptions or empirical relations.



(a) **(b)**

Figure 6: Evaluation of uncertainty quantification method where (a) shows the bootstrapped distribution of x-displacements for a single synthetic video with 1000 iterations and (b) shows the distributions of studentised residual for the synthetic videos. The red vertical lines in (a) show the standard deviation of the distribution giving an estimate of the standard error. The red curve in (b) shows the expected distribution of the studentised residuals.

The only assumption made in the method is that the set of correlation matrices are representative of the underlying population. Additionally, the accuracy of the estimate of the standard error can be improved further by applying more iterations of the bootstrapping technique.

4.3. Evaluation of correlation averaging and phase binning

Figure 7 shows the relationship between averaging non-correlation averaged measurements and using correlation-averaging. If both methods were equivalent we would see a clustering around the red line. The fact that the distribution is displaced from the red line indicates that the there is a systematic bias in one of the methods. This bias is likely a zero-bias in the non-correlation-averaged measurements due to a significant number of noisy frames. These frames generally produce a uniform distribution of velocities which, when added to the normally distributed non-noisy frames, gives a zero-bias. Figure 8a shows the relationship between the correlation-averaged mean and the mean phase. Figure 8a shows a symmetrical distribution along the central phase suggesting that there is limited velocity variation across the phase bin. This suggests that the phase-bin size is sufficiently small. Likewise Figure 8b shows limited evidence for any correlation between timestamp and the displacement vector. This suggests the technique used to assign phases to the SPIM video performs well with limited or no phase-drifting.

4.4. Identifying beat-to-beat variability within the zebrafish heart

Detecting beat-to-beat variability is a very difficult problem as it is difficult to separate the inherent variability within the data with the uncertainty induced through measurement error. This is compounded by the correlation-averaged technique which was applied in this project which means that the approaches applied here had to rely on the bootstrapped distribution.

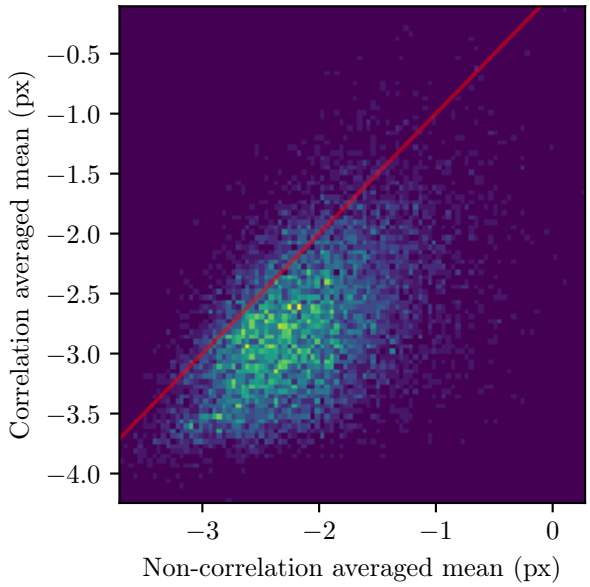


Figure 7: Bootstrapped correlation between correlation-averaged mean and non-correlation-averaged mean. The red line shows the expected location and angle of the distribution if both methods produced the same result. The non-correlation-averaged mean is biased to lower estimates of the displacement.

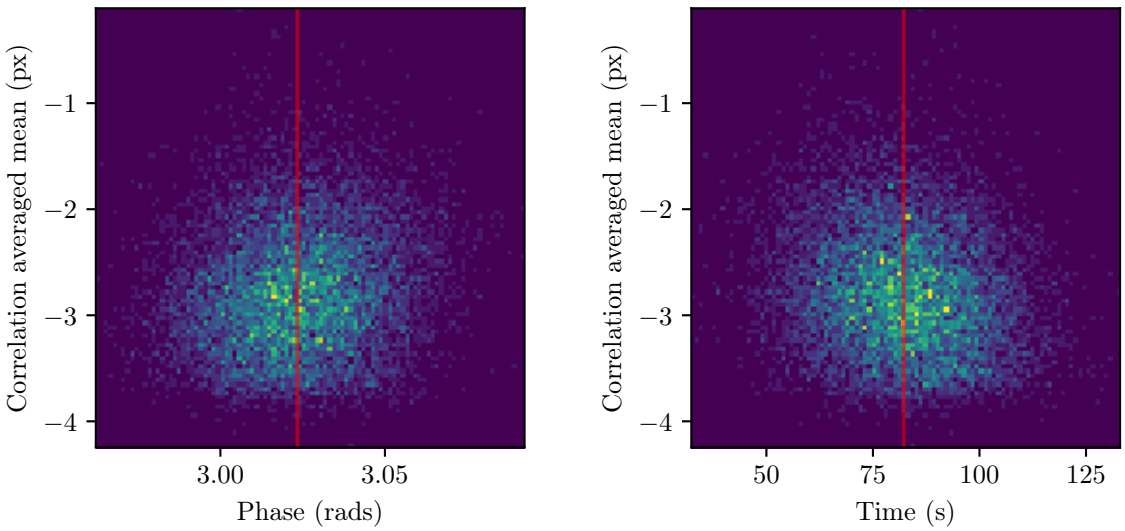


Figure 8: Bootstrap correlations between correlation-averaged displacement and (a) phase and (b) timestamp. The red lines show the mean (a) phase and (b) timestamp. The symmetry and lack of skew indicate limited correlation between the measurements.

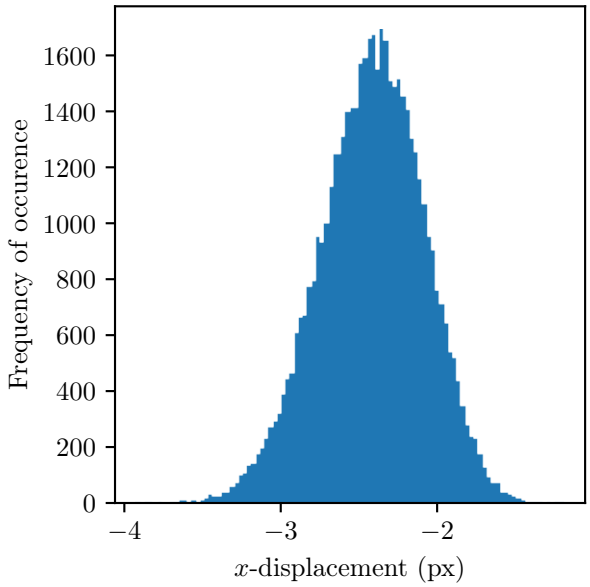


Figure 9: Bootstrapped x-displacement distribution in the zebrafish heart.

4.4.1. Bootstrapping

While Figure 9 shows no clear evidence of multimodality suggesting that there is no significant beat-to-beat variability, the synthetic images with known outliers also do not show multimodality. This suggests that the presence of outliers does not induce multimodality in the bootstrap distribution and this is therefore not an effective method for identifying beat-to-beat variability. This matches what was found by Singh and Xie [19].

The synthetic videos showed increased skew and kurtosis in the presence of outliers with a skew of 0.27 ± 0.01 and a kurtosis of 0.13 ± 0.03 compared to the synthetic video without outliers with a skew of 0.19 ± 0.01 and a kurtosis of 0.06 ± 0.02. The p-value for the hypothesis that the videos with outliers have a greater skew and kurtosis was 1.15×10^{-5} and 0.004 respectively. These p-values provide very strong evidence that the presence of outliers is associated with increased skew and kurtosis. The skew and kurtosis of the zebrafish distribution was close to that of the synthetic videos without outliers indicating that there is limited beat-to-beat variability in the zebrafish heart. It is unknown, however, if the skew and kurtosis of this distribution is influenced by other factors in the images such as: seeding density, levels of noise, and spacial and temporal velocity gradients. Further work is needed to investigate this.

We might intuitively expect that outliers would result in a negative kurtosis, indicating greater spread in the data, but surprisingly, the synthetic data showed that the data with outliers has a positive kurtosis. This kurtosis was greater than what was found in the synthetic video without outliers. The reason for this is unknown and requires further investigation.

4.4.2. Bootlier

The zebrafish bootlier plot in Figure 10 is uni-modal which would suggest there are no outliers in the zebrafish data. Looking at Figure 11, however, shows that the method did not detect outliers in the synthetic data which was known to contain outliers suggesting this is not a good approach. The approach likely does not work due to the use of correlation-averaging. If the correlation-averaged peak's signal is located reasonably far away from the secondary peak then

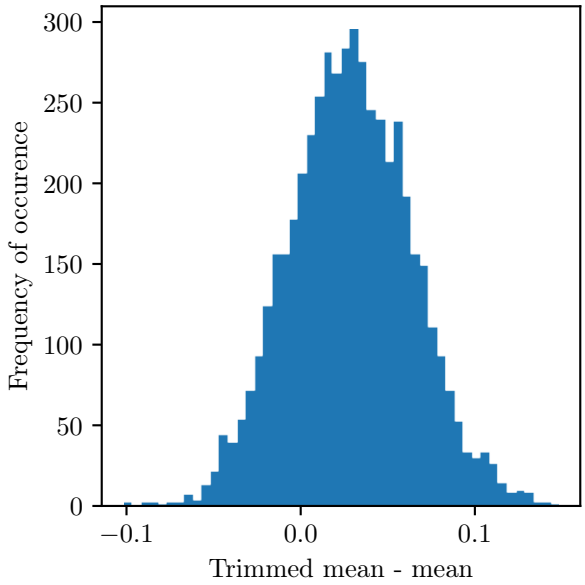
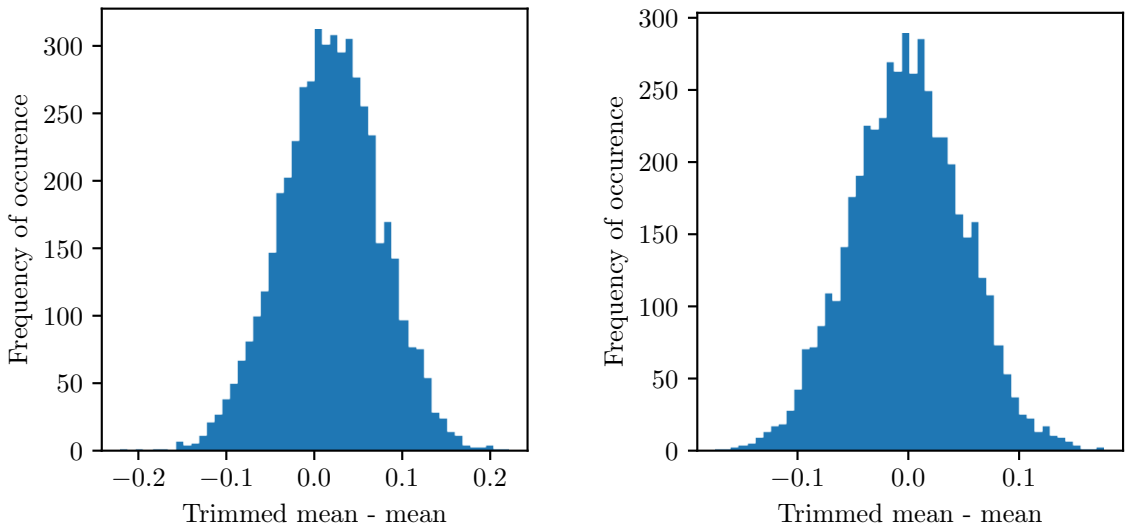


Figure 10: Bootlier x-displacement distribution in the zebrafish heart. No multimodality is visible.



(a) **(b)**

Figure 11: Bootlier plots for synthetic videos with (a) known outliers and (b) no outliers. No multimodality is visible in either plot.

the secondary peak will have a minimal impact on the estimated displacement. The approach may perform better on the non-correlation averaged data but it is possible that the presence of noise frames in the non-correlation averaged data would produce false-outliers which do not truly represent the flow of particles.

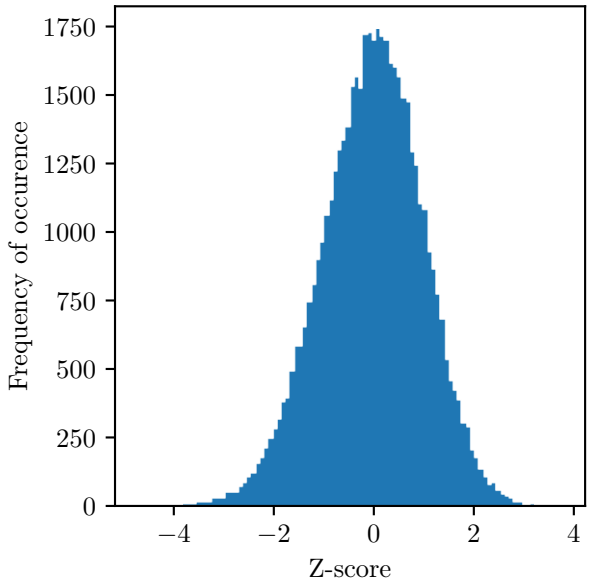


Figure 12: Modified z-score applied to the zebrafish heart's bootstrapped distribution. Values lying above 3.5 or below −3.5 indicate the presence of outliers.

4.4.3. Modified Z-score

Figure 12 shows the modified z-scores for the zebrafish heart. There were 2 measurements with modified z-scores less than −3.5 and 31 greater than 3.5. The synthetic video with outliers had a mean number of z-scores greater than 3.5 of 0.55 ± 0.03 and the video without outliers had a mean number of z-scores greater than 3.5 of 0.35 ± 0.03. The p-value that the video without outliers had a greater number of modified z-scores greater than the video without outliers was 8.57×10^{-6} providing good evidence that the modified z-score is correlated with the presence of outliers. Approximately 0.27% of bootstrapped measurements in the synthetic video with outliers contained modified z-scores greater than 3.5 compared to the video without outliers which had approximately 0.17%. The zebrafish video had only 0.06% of its measurements greater than 3.5 and 0.004% less than −3.5. We can form a few conclusions from this. Firstly, this is smaller than either of the sets of synthetic videos, suggesting that there are limited outliers but also that other factors may influence the modified z-scores. Secondly, if there are outliers, it provides evidence that these lie on the left side of the distribution indicating the outliers are negative displacements.

4.4.4. Window-transformation

Manually comparing the window-transformed frames to the true frames shows strong agreement with the velocities within the ventricle and the atrium. The valve region, however, showed some velocity underestimation. This is likely partially due to significant velocity shear and gradients within this region. The impact of these velocity shear and gradients on measurements could be partially reduced by implementing the window-deformation PIV method which performs substantially better under these conditions [24].

4.5. Future work

It is difficult to directly verify the measurements taken using this method. The results do show similarities to the work done by Zickus and Taylor [3] where similar methods were applied to a

different data set. The window deformation technique provided a good method for verification of the measurements and the method could be improved by the application of local contrast normalisation. This would allow for the predicted frame to be directly compared to the expected frame using a similarity index such as the sum of absolute differences. Another method to verify the results would be to use particle-tracking velocimetry. This approach directly tracks individual particles. The difficulty with this technique is accurately detecting particles between frames. A combined approach applying PIV and PTV could simplify this as the PIV system could help identify matching particles.

Identifying variability within the data is a difficult problem and the methods presented had mixed success. This is partly due to the difficulty in disassociating the variability introduced due to velocity gradients across a phase bin. A solution to this would be to fit a linear model to the correlation in Figure 8a. Corrections to the bootstrapped velocity distributions could then be applied. This would, however, introduce an additional uncertainty and, from Figure 8a, we can see that the correlation is weak and therefore the additional uncertainty may be greater than the variation introduced from the phase-binning. It was also difficult to identify whether any variation identified was due to noise or true variability. If a robust measure of noise was identified then applying the measure to the outliers identified using the modified z-score method would provide an indication if they were true variation or just the result of statistical noise.

It would be useful to apply these approaches to a zebrafish heart with known variability. For example, a zebrafish with atrioventricular block. This would allow direct comparison between the methods applied here on a zebrafish which we have assumed has minimal flow velocity variation. This avoids the need for synthetic images which do not accurately represent the conditions found in the SPIM videos.

5. Conclusion

In this work a PIV analysis pipeline was developed and applied to the zebrafish heart to measure blood flow velocities, reproducing the work of Zickus and Taylor [3]. The project set out to build upon their work by developing new methods to quantify the uncertainties on their measurements and to evaluate the validity of their key assumptions. A new technique was successfully developed, requiring few assumptions, to quantify the uncertainties on correlation-averaged PIV measurements. The technique was found to perform well on simulated data, generally producing only a small overestimate of the uncertainty. However, a small number of significant underestimates were identified. Application of the technique to the zebrafish heart indicated that the uncertainties are relatively small but grow significantly in regions with high flow velocities. The uncertainty quantification technique was extended to evaluate the phase-binning method which was shown to perform well, with minimal variation of measured flow velocities within the phase-bin under investigation and no noticeable variation over time. Attempts to identify beat-to-beat variability found mixed success. The modified z-score method and the bootstrapping approach showed some promise, with good evidence that they are sensitive to outliers which would indicate variability. Applying these techniques to the zebrafish heart showed little evidence of significant beat-to-beat variability in the region of interest providing good evidence that this key assumption of the technique is valid. Future research could build upon these approaches, by applying them to biological models which are known to have some beat-to-beat variability.

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